



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Affiliated to DBATU, RTMNU & MSBTE, Mumbai
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

An Autonomous Institute, with NAAC "A" Grade

Department of Computer Science & Engineering/Artificial Intelligence

"A place to learn; A chance to grow"

Session: 2025-26



MID SEM EXAMINATION- I

Course: B. Tech. in CSE/AI

Year/Semester: 3rd/5th Semester

Subject: Ethics, Bias and Safety in LLM

Subject Code: CS5M004

Max Marks: 20

Date: 13/08/2025

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Analyze ethical challenges in LLM development and deployment.
2. Detect and mitigate bias in training datasets and model outputs.
3. Assess and improve the safety of LLM-based applications
4. Understand legal and regulatory frameworks surrounding AI usage.
5. Evaluate and propose solutions to ethical controversies in LLMs using real-world case studies.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define AI ethics.	L1/CO1	1
Q.2	State principle of safety.	L1/CO1	1
Q.3	Exemplify sentence contradiction in LLM.	L1/CO2	1
Q.4	List the strategies for mitigating bias in AI.	L1/CO2	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Describe the concept of transparency in LLMs.	L2/CO1	2
Q.6	Summarize the role of datasets in introducing bias.	L2/CO1	2
Q.7	Justify the importance of explainability in AI systems.	L2/CO2	2
Q.8	Elaborate one method to mitigate biases in LLMs.	L2/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Illustrate with examples the ethical issues that may arise in LLM deployment in healthcare or education.	L3/CO1	4
Q.10	Propose a framework for ensuring fairness in LLM development in healthcare.	L3/CO1	4
Q.11	An Actor realized unethical use of his images for reel making by others; Propose a solution for prohibiting this.	L3/CO2	4